Group 3 Activity Packet — Digital Watermarking

"The Art Ownership Dispute" — self-run, Judge-led, no teacher intervention needed

How This Document Works

This packet is designed so your group can run the entire activity yourselves, without waiting for the teacher at any point. Follow the stages below IN ORDER — each stage tells you exactly what you receive, what to do with it, and who to hand your result to next.

- The teacher will not check your work step-by-step during the activity. You are expected to self-check as you go (see the yellow boxes) and cross-check with your teammate(s) (see the blue boxes).
- Where you see a blank line, that is YOUR job to work out — the answer is deliberately not printed anywhere in this document. If you get stuck, re-read the relevant section of your Study Material.
- The Judge / Timekeeper (the last stage) is in charge of running your group's clock, confirming the final results match, scoring part of the rubric themselves using the guidance printed in their own stage, and filling in the Result Card at the end — that Result Card is what gets submitted for grading, and the teacher's script totals it automatically.

Do not skip ahead: Only read the stage that applies to your own role. Reading a later stage's instructions before it's your turn defeats the purpose of the activity — you'll be found out anyway, because the numbers won't make sense to you without the steps in between.

The Scenario

A digital artist wants to be able to prove authorship of an artwork even after it has been copied and republished by someone else. Before publishing, they embed a hidden 4-bit ownership mark into four specific pixels, chosen by a secret key, and register that key with a trusted third party. Later, a rival publishes the same artwork and falsely claims to have created it. A Verifier team must use the registered key to determine, beyond doubt, who the real creator is.

Roles in Your Group (9 students)

- Students 1, 2 — Owner / Creator A & B Anitta/Filu
- Student 3 — Registrar Mathew
- Student 4 — Publisher / Channel Joel
- Students 5, 6 — Claimant A & B Ananya/Johan
- Students 7, 8 — Verifier A & B Alby/Dyuthi
- Student 9 — Judge / Timekeeper Nigel

Given Data (the only numbers you start with)

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Artwork (16 pixel values):
[200, 198, 205, 210, 207, 199, 203, 208, 206, 201, 204, 209, 202, 197, 211, 205]
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The key, the watermark positions, and the watermarked artwork must all be worked out by the roles below — none of it is given upfront. Your instructor will privately give the Owner team a 4-bit watermark pattern and a key to use.

Stage-by-Stage Instructions

Stage 1 — Owner / Creator (Students 1 & 2)

You start with: The artwork array from the Given Data, plus a secret key and a 4-bit watermark pattern given privately to you by your instructor before the activity begins.

What to do:

1. Using the formula position(j) = (key × j + 1) mod 16 for j = 0, 1, 2, 3, compute the four pixel positions where your watermark will go.

13 = 16 x 0 + 13

1111

2. Overwrite the LSB of the artwork's pixel value at exactly those 4 positions with your 4 watermark bits, in order.

Self-check before you hand off: Before handing your watermarked artwork to the Publisher, extract the LSBs back out at your own 4 positions and confirm they exactly match the watermark bits you were given. If they don't, you've made an embedding slip — find and fix it now, since nobody downstream can catch this for you.

Cross-check with your partner(s): Both of you compute the 4 positions and the watermarked pixel values independently, then compare all values before moving on.

Your 4 positions and the final watermarked artwork array (all 16 values):

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Hand off: First, privately give the Registrar your key and true watermark bits — this MUST happen before you give anything to the Publisher, and must stay hidden from the Claimant team. Only after that, hand the finished watermarked artwork to the Publisher to the Registrar (privately) and then the Publisher. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 2 — Registrar (Student 3)

You receive: The secret key and the true watermark bits, given to you privately before publication. — handed to you by Owner.

What to do:

3. Seal and hold this information — do not share it with anyone, including the Claimant team, unless a dispute is formally raised later by the Judge.
4. When the Judge asks you for it during the Verifier's stage, release the key and true bits ONLY to the Verifier team.

Hand off: Release the sealed key and watermark bits to the Verifier team, later, only when asked. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 3 — Publisher / Channel (Student 4)

You receive: The finished watermarked artwork. — handed to you by Owner.

What to do:

5. Make copies of the watermarked artwork available to both the Claimant team and the Verifier team — exactly as a public gallery or website would.

Hand off: Distribute copies of the watermarked artwork to the Claimant team and Verifier team, both. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 4 — Claimant — attack path (Students 5 & 6)

You receive: A published copy of the watermarked artwork — no key. — handed to you by Publisher.

What to do:

6. Falsely announce that you created this artwork.
7. Try each candidate key from {2, 4, 6, 8}: for each one, compute its 4 positions using the SAME formula the Owner used (position(j) = key×j+1 mod 16), then extract the LSBs at those positions.
8. For each key, note whether the resulting 4-bit pattern looks like a plausible watermark, or is invalid (e.g. a key that produces a repeated position), or just doesn't look meaningful.
9. Report which key(s), if any, produced a pattern that could support your false claim.

Cross-check with your partner(s): Both of you compute each candidate key's positions and extracted bits independently, then compare before reporting.

Your results for keys 2, 4, 6, and 8 (positions + extracted bits for each):

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Hand off: Report your findings for all 4 candidate keys to the Judge. Do not show them this stage's instructions — only give them the value(s) named above.

Verifier — legitimate

eceive: The registered key a
as both the Owner's and the Cl
sealed record from the Registrar o

What to do:

10. Using the registered key, co
11. Extract the LSBs at thos

watermark bits.

pixel values

tract the LSBs back out
ney don't, you've made

— Verifier — legitimate path (Students 7 & 8)

receive: The registered key and true watermark bits (released by the Registrar once the Judge raises the dispute), us both the Owner's and the Claimant's copies of the artwork. — handed to you by the Judge (who requests the sealed record from the Registrar on your behalf) and the Publisher.

What to do:

10. Using the registered key, compute the 4 registered positions.
11. Extract the LSBs at those exact positions from the OWNER's copy of the artwork.
12. Extract the LSBs at those exact same positions from the CLAIMANT's copy of the artwork.
13. Compare both results against the true registered watermark bits.
14. Record how long this verification took you, in minutes.

Cross-check with your partner(s): Both of you extract the bits from both copies independently, then compare all four sets of bits (yours and your partner's, for both copies) before reporting.

Bits extracted from Owner's copy, bits extracted from Claimant's copy, and your ruling:

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Hand off: Report which copy (if either) matches the true registered watermark, and your time taken to the Judge. Do not show them this stage's instructions — only give them the value(s) named above.

Stage 6 — Judge / Timekeeper — you lead this group (Student 9)

You receive: The Claimant's candidate-key findings and the Verifier's extraction results from both copies. — handed to you by Claimant team and Verifier team.

What to do:

15. You are in charge of running this activity for your group — keep time for each stage above and keep everyone moving, without waiting for the teacher.
16. Once the Owner's stage is done, formally raise the dispute: ask the Registrar to release the sealed key and bits to the Verifier team.
17. Read the Verifier's ruling: which copy matches the true registered watermark?
18. Announce the final decision to the group: ownership is confirmed for whichever copy matched.

Score before you fill in the Result Card: As Judge, you also score three rows of the rubric yourself, right now, using the guidance below — there is no separate teacher check for these.

19. **Attacker/Claimant correctness (0-5):** full 5 if the Claimant team correctly showed NO valid key reproduces the TRUE registered watermark (i.e. their forgery attempt genuinely failed and they reported that honestly, including flagging any coincidental match like key=2 for what it is); partial 2-3 if they used the right method but made an error or ran out of time.
20. **Attacker/Claimant efficiency (0-3):** reward a systematic method over guessing — e.g. correctly distinguishing a real watermark match from a coincidental one.
21. **Explanation quality (0-3):** before scoring this, pick ONE member at random from each sub-team (not the strongest student) and ask them to explain one step out loud. Full marks only if they can actually explain it, not just read out a number.

Once scored, fill in the Result Card on the next page completely and legibly.

Result Card — fill this in and submit

The Judge / Timekeeper fills in this card once both the legitimate path and the attack/claim path are complete confirmed to match, or the Verifier's ruling is complete. This card is what your teacher will use to grade your g automatically — fill in every field exactly as instructed, using digits/letters only where shown.

GROUP: 3

STUDENT NAMES / ROLL NUMBERS (all 9, by role):

LEGIT_RESULT (Verifier's ruling — write OWNER or CLAIMANT):

LEGIT_TIME_MIN (Verifier's time, in minutes): 5min 9sec

OWNER_BITS (4 bits Verifier extracted from Owner's copy):
1010

CLAIMANT_BITS (4 bits Verifier extracted from Claimant's copy):
1010

ATTACK_RESULT (best-matching candidate key the Claimant found, or NONE):
Yes the claimant key matched the key generated by the owner and the key has been verified by the verifier

ATTACK_METHOD_NOTE (1-2 lines: did any candidate key coincidentally match? how did you tell it wasn't the real key?): No

ATTACK_RESULT_SCORE (0-5, using the correctness guidance above):
5 (The claimaint correctly guessed the key)

ATTACK_EFFICIENCY_SCORE (0-3, using the efficiency guidance above):
~~5~~ 3

EXPLANATION_SCORE (0-3, from the random member you asked to explain a step):
~~4~~ 3

JUDGE_SUMMARY (1 line: whose favour was the ruling, and why?):
The claimant ~~g~~ checked the 4 keys they were provided and correctly guessed the key which was 6 that was created by the owner.

How to submit: Hand this page (or a clear photo of it) to your teacher. If asked to submit digitally, type these exact values into a plain text file named Group3_Result.txt, one "FIELD: value" line per field exactly as labelled above, and send it as your teacher instructs.

Overall Time : ~~45min 12sec~~ 35min 18sec

Student Name	Roll number
~~Vari~~Filu Minta Tom	12 / Owner
Nigel James Noby	25 / Judge.
Dyuthis S Gareth	11 / Verifier
Alby Mathey Biju	3 / Verifier
Johan P Mathew	17 / Claimant
Mathew Manoj	23 / Registrar
Joel S Joseph.	16~~8~~ / Publisher
Anitta Vincent	5 / Owner
Aranya R Nair	4 / Owner Claimant

Score The 3 Judge

...nree fields are the only on ...d otherwise apply. Fill these

ATTACK_RESULT_SCORE (0-5)
Full 5 if the Claimant team correc attempt genuinely failed and they Partial 2-3 if they used the right me

ATTACK_EFFICIENCY_SCORE (0-3)
Reward a systematic method over g rather than treating any matching ... or skipped ...

path are complete
use to grade your grou

Score The 3 Judgement Rows

three fields are the only ones where YOU decide the number, using the guidance below — the same guidance a teacher uld otherwise apply. Fill these in before you touch the Result Card.

ATTACK_RESULT_SCORE (0-5)

Full 5 if the Claimant team correctly showed NO valid key reproduces the TRUE registered watermark — i.e. their forgery attempt genuinely failed and they reported that honestly, including flagging any coincidental pattern match for what it is. Partial 2-3 if they used the right method but made an error or ran out of time.

ATTACK_EFFICIENCY_SCORE (0-3)

Reward a systematic method over guessing — e.g. correctly distinguishing a real watermark match from a coincidental one, rather than treating any matching pattern as proof. Give 3 for a clearly systematic approach, 1-2 if they got there haphazardly or skipped a candidate they should have checked, 0 for no real method.

EXPLANATION_SCORE (0-3)

Before writing this score, pick ONE member at random from each sub-team — not the strongest student — and ask them to explain one step out loud. Score 3 only if they can actually explain it (not just read out a number), 1-2 for a partial explanation, 0 if they can't explain it at all.

Do this before you write the number: The explanation check has to happen live, right now, with a student you pick — not the team's own choice of who answers.

Submitting

How to submit: Hand the completed Result Card page (or a clear photo of it) to your teacher. If asked to submit digitally, type the exact values into a plain text file named Group3_Result.txt, one "FIELD: value" line per field, exactly as labelled above.